



$$f(x) = x^3 - 3ax^2 + 3a^2x - 3|x| + 3$$

2 экстр

$(-2, 3)$

$$f'(x) = 3x^2 - 6ax + 3a^2 - 3$$

$$1) x \geq 0$$

$$x \geq 0$$

$$f'(x) = 3x^2 - 6ax + 3a^2 - 3 = 0$$

$$1) D < 0$$

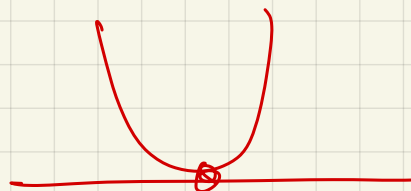
$\ominus$



0 экстр

$$2) D = 0$$

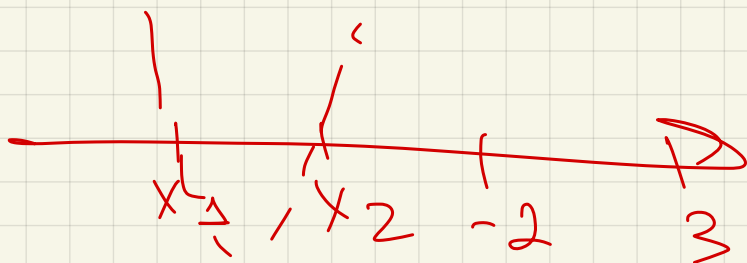
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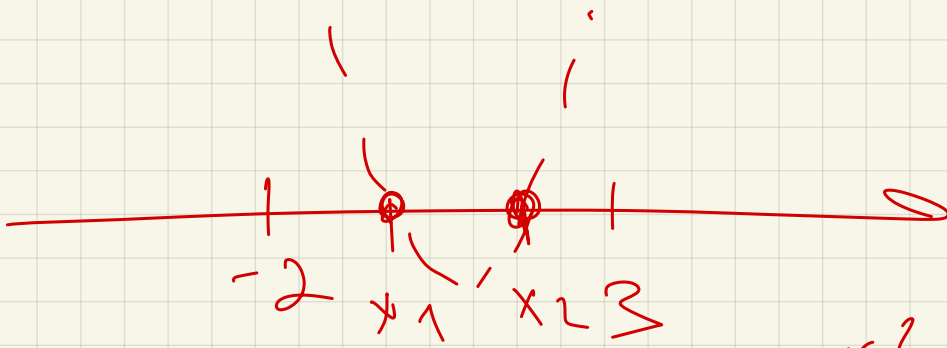
$$3) D > 0$$

$$x_1 =$$

$$x_2 =$$



$\emptyset$



$$\begin{cases} x_2 < 3 \\ x_1 > -2 \end{cases}$$

$$x^2 - 2cx + c^2 + 1$$

$$\frac{2c \pm \sqrt{4c^2 - 4(c^2 + 1)}}{2}$$

$$2c \pm \sqrt{4c^2 - 4c^2 - 4}$$

$$\Delta < 0$$

$$\frac{2c \pm 2}{2} = c \pm 1$$

$$\frac{a+1}{a-1}$$

$$a = 100$$

$$a = -100$$

$$ax^2 + 2x - 2,25 = 0$$

$$f(x) = 2x^3 - 9x^2 - 6ax + 13,2$$

$$|x_1 - x_2| = \sqrt{(x_1 - x_2)^2} =$$

$$\sqrt{(x_1 + x_2)^2 - 4x_1 \cdot x_2} =$$

$$x_1 + x_2 = -\frac{2}{a}$$

$$x_1 \cdot x_2 = -\frac{9}{4a}$$

$$\sqrt{\left(-\frac{2}{a}\right)^2 - 4\left(-\frac{9}{4a}\right)} = \sqrt{\frac{4}{a^2} + \frac{9}{a}} =$$

$$\sqrt{\frac{4 + 9a}{a^2}}$$

$$f'(x) = 6x^2 - 18x - 6a = 0$$

$$6(x^2 - 3x - a) = 0$$

$$x_3 \quad x_4$$

$$|x_3 - x_4|$$

$$x_{3,4} = \frac{3 \pm \sqrt{9+4a}}{2}$$

$$\frac{3 + \sqrt{9+4a}}{2}$$

$$\frac{3 - \sqrt{9+4a}}{2}$$

$$\left| \frac{3 + \sqrt{9+4a}}{2} - \left(\frac{3 - \sqrt{9+4a}}{2} \right) \right| =$$

$$\frac{3 + \sqrt{9+4a} - 3 + \sqrt{9+4a}}{2} = \frac{2\sqrt{9+4a}}{2}$$

$$= \sqrt{9+4a}$$

$$\sqrt{\frac{4+9a}{a^2}} \leq \sqrt{9+4a}$$

$$\frac{4+9a}{a^2} \leq 9+4a$$

$$a > 0$$

$$4 + a \leq 9a^2 + 4a^3$$

$$4a^3 + 9a^2 - 9a - 4 \geq 0$$

$$a = 1$$

$$\begin{array}{r} 4a^3 + 9a^2 - 9a - 4 \big| a-1 \\ \underline{4a^3 - 4a^2} \\ 13a^2 - 9a - 4 \\ \underline{-13a^2 + 13a} \\ 4a - 4 \end{array}$$

$$(4a^2 + 13a + 4)(a-1) \geq 0$$

$$13 \pm \sqrt{13^2 - 4 \cdot 4 \cdot 4}$$



a

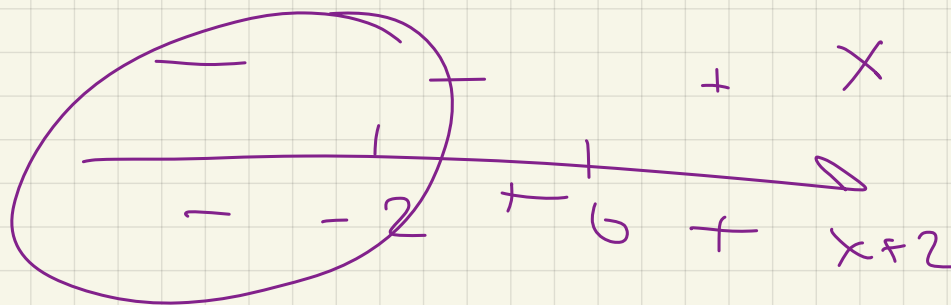
$$\frac{x+2}{|x+2|} + |x|(x^2-48) = a$$

3 plu

$$x \neq -2$$

$$-1 - x(x^2-48) = a$$

$$x < -2$$



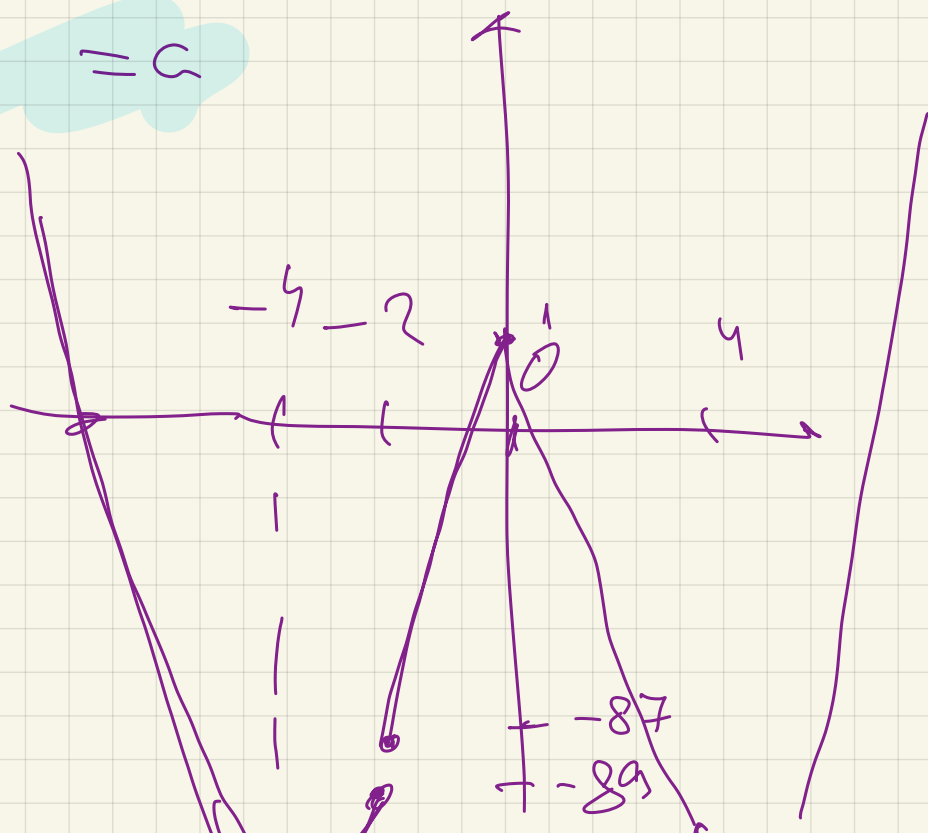
$$-1 - x|x^2-48| = a$$

$$-x^3 + 48x - 1 = a$$

$$8 - 48 \cdot 2 - 1$$

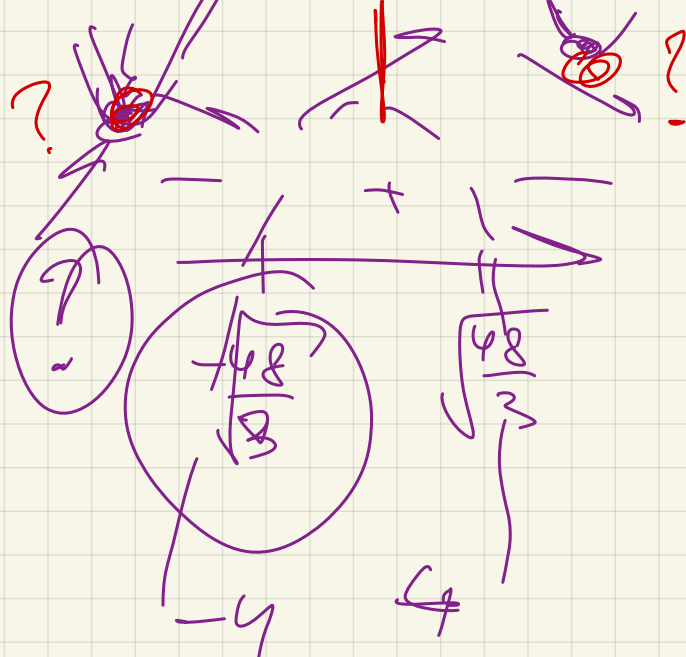
$$-x^3 + 48x + 1 = a$$

$$-3x^2 + 48$$



$$f'(x) = -3x^2 + 48$$

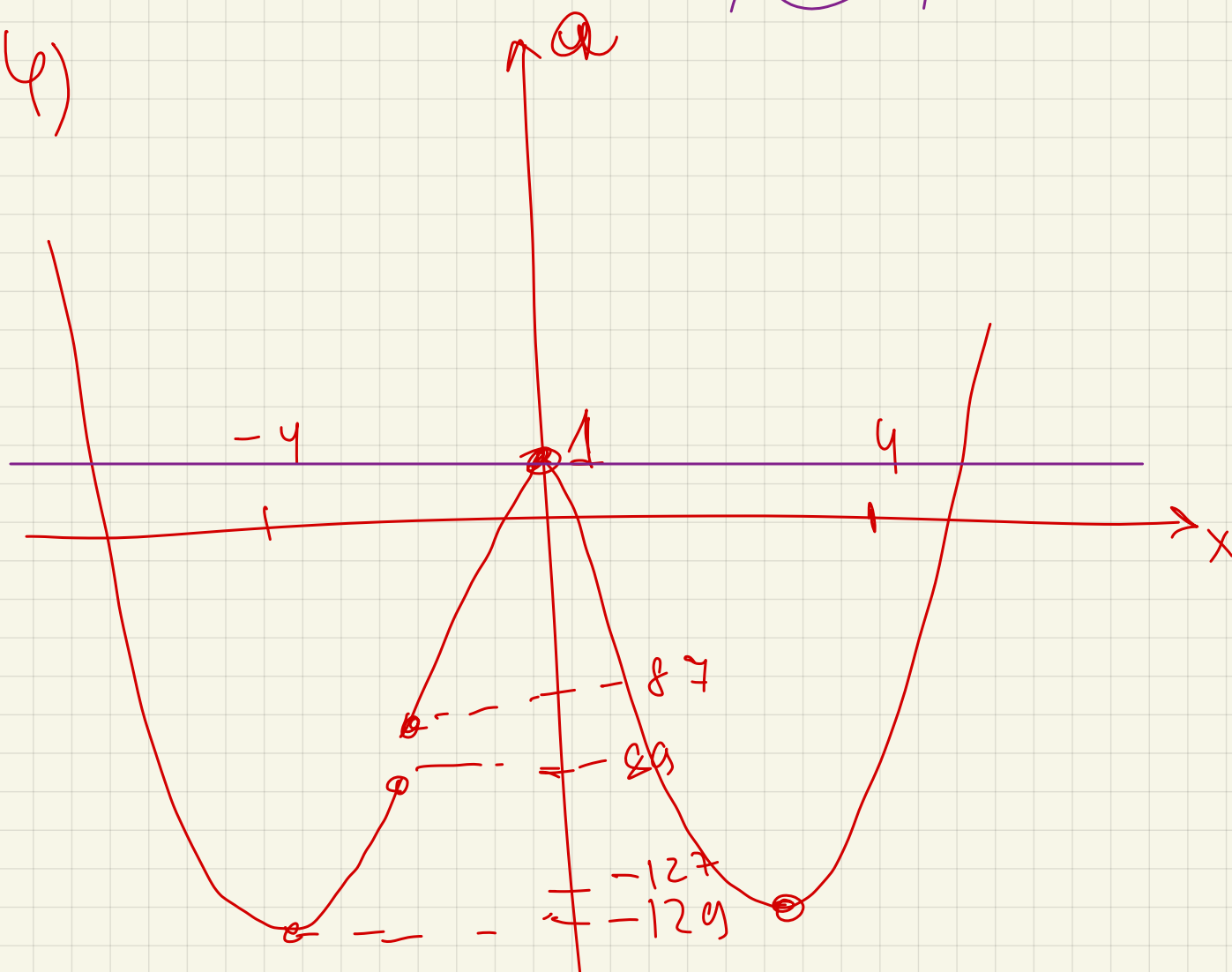
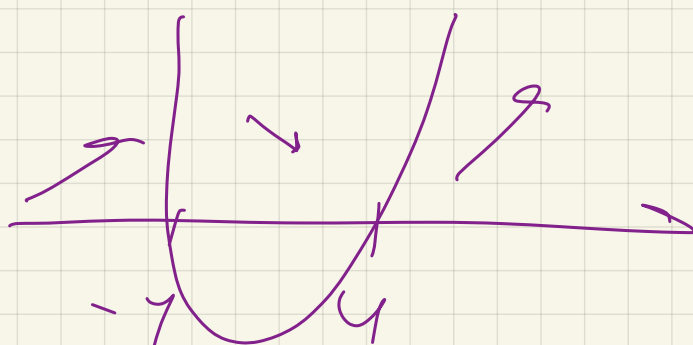
$$x^2 = \frac{48}{3}$$



$$+8 - 96 + 1$$

$$8x^2 - 48 + 1$$

(4)



$$a = -127$$

$$Q \in [-89 : -87]$$

$$a = 1$$